DEMKit Microgrid - Quick Start for Teachers

Status: internal classroom evaluation guide (Windows prototype).

This is a short guide for trying the simulation with students and giving feedback. The Windows desktop prototype works on the developer's computer, but it has not been approved for general distribution. Installation on a clean computer without Python or Node.js, uninstall behaviour, dependency security and redistribution rights still need checking. Use only an installation supplied through an approved internal test.

What is the tool?

The simulator shows a small neighbourhood with six houses connected to an electricity grid. Students can try EV chargers, solar panels, batteries and household devices, then see how the electricity demand and local production change.

The point is to experiment and discuss, not to predict exact electricity costs or the real capacity of a particular neighbourhood.

Open the simulation

1. On an approved test computer, open DEMKit Microgrid Prototype from the Windows desktop or Start menu.
2. Wait until the game appears. You do not need to open a separate Python or Node.js terminal.
3. Read the introduction and click Start Mission.
4. The simulation shows the houses, a clock, transformer reading, device choices and a coin budget.

The Electron window can be smaller than the full 1920 x 1080 game board. Use the mouse wheel, touchpad or scrollbars to see the lower houses or the right side. This scrolling change is part of the latest development code; the installed version must be rebuilt after that change to include it.

If the program shows a startup error, stop and report the error message to the project developer. Don't install extra software or dismiss security warnings just to make it work.

Try a simple activity (10-15 minutes)

1. Start with one house. Look at its current load and power.
2. Add an EV charger. Drag the EV icon from the device area into a house. See whether its power and the coin budget change.
3. Add solar panels. Compare the house's reading with and without solar production. Output can depend on the simulated time of day.
4. Add a battery. Try the Solar, Smart, plus and minus controls. A battery does not always change immediately: it depends on charging state and demand.
5. Check the goals. Read the current goal and complete it. Note any coin reward.
6. Look at the grid. Compare load and power between houses and watch for warnings. An overload may not happen in every short session.

The displayed device prices include EV charger: 10 coins, solar panel: 10 coins, battery: 30 coins in the tested game version. Coins, house slots and goals are game mechanics, not real purchase prices.

Things to look at while playing

- House load % and Power W: changes when devices use or produce electricity.
- Transformer A: a reading for the shared electricity network.
- Coins and house slots: limits on what students can buy and place in the game.
- Battery modes: charging/discharging behaviour may vary with demand and solar.
- Overload warning: something to discuss about network capacity; don't force a failure if it doesn't appear.
- Time and season: help explain why demand or solar generation changes.

Discussion after the activity (5 minutes)

Ask students:

1. What happened to the grid when we added EV chargers or other devices?
2. When did solar panels or batteries help, and when were they less useful?
3. Who could afford these technologies in a real neighbourhood, and who might be left out?
4. Who would pay for a stronger electricity network if it was overloaded?
5. What important real-world details are missing from this simple game?

The last three questions connect the exercise to fairness in the energy transition. The game's equal starting budget does not mean all real households have equal money, housing choices or access to technology.

At the end

Close the desktop application normally. The latest prototype has been tested reopening on the original Windows computer, but saving progress across new launches is not a promised feature. Treat each lesson as a new test session until reset/persistence is checked.

If something goes wrong

Problem	What to do
Only part of the neighbourhood is visible	Use the scrollbars or mouse wheel; maximize the window if useful.
A device won't fit in a house	Check available slots, budget and whether the game allows that action.
Values don't change right away	Check whether the simulation is running, and whether the simulated conditions allow that device to act.
App says it cannot start	Copy the exact message and send it to the developer.
Web server disconnected	Close the application, make sure an older copy is not running and try once more. Report repeated failures.
Windows shows an unknown publisher or security block	Stop and contact the supervisor/developer. Do not bypass a security block.

Feedback for the project team

After a small supervised test, note: How long did starting the app take? Could students reach all six houses? Could they drag EV/solar/battery devices? Did the explanation make sense? What confused them? Which new feature would help a lesson most?

Project: DEMKit Microgrid Touchtable - Saxion educational prototype. For technical setup and known limitations, see the developer guide.